Faculty: FTMI
Course: Введение в веб технологии
Year: 2025/2026
Group: U4225
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Лабораторная работа №1

Ход работы

1. Установлен Докер

```
Docker version 29.2.1, build a5c7197
flocktory@MacBook-Air-Flocktory-69 ~ % docker --version
Docker version 29.2.1, build a5c7197
```

```
flocktory@MacBook-Air-Flocktory-69 ~ % docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
198f93fd5094: Pull complete
95ce02e4a4f1: Download complete
Digest: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9dbdd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (arm64v8)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
```

```
flocktory@MacBook-Air-Flocktory-69 ~ % docker images
                                                    i Info →   U  In Us
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   85404b3c5395       22.5kB         10.2kB    U
```

```
flocktory@MacBook-Air-Flocktory-69 ~ % docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
flocktory@MacBook-Air-Flocktory-69 ~ % docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED              STATUS                          PORTS     NAMES
2c08209d64b3   hello-world   "/hello"   About a minute ago   Exited (0) About a minute ago             eager_swirles
```

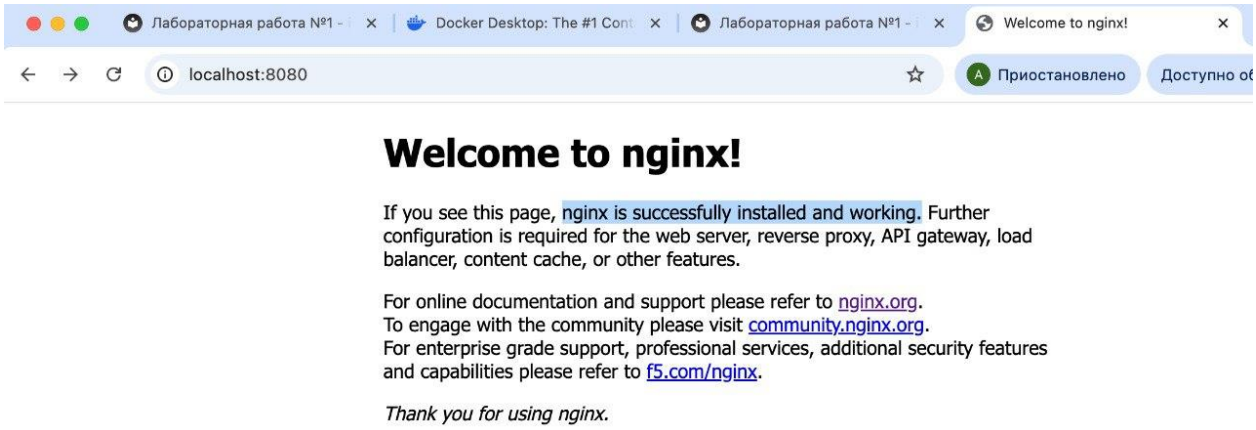
2. Работа с готовыми образами

```
CONTAINER ID   IMAGE          COMMAND    CREATED              STATUS                          PORTS     NAMES
2c08209d64b3   hello-world    "/hello"   About a minute ago   Exited (0) About a minute ago             eager_swirles
flocktory@MacBook-Air-Flocktory-69 ~ % docker pull ubuntu:latest
latest: Pulling from library/ubuntu
66a4bbbfab88: Pull complete
9c2a2ec78563: Download complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
flocktory@MacBook-Air-Flocktory-69 ~ % docker run -it ubuntu bash
root@b8dc6d541e31:/# apt update && apt install -y curl
Get:1 http://ports.ubuntu.com/ubuntu-ports noble InRelease [256 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates InRelease [126 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-backports InRelease [126 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-security InRelease [126 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 Packages [19.0 MB]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 Packages [1776 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble/restricted arm64 Packages [113 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble/multiverse arm64 Packages [274 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse arm64 Packages [41.7 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 Packages [2390 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 Packages [1977 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted arm64 Packages [4825 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports noble-backports/universe arm64 Packages [34.7 kB]
Get:14 http://ports.ubuntu.com/ubuntu-ports noble-backports/main arm64 Packages [49.4 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports noble-security/universe arm64 Packages [1262 kB]
Get:16 http://ports.ubuntu.com/ubuntu-ports noble-security/multiverse arm64 Packages [42.1 kB]
Get:17 http://ports.ubuntu.com/ubuntu-ports noble-security/restricted arm64 Packages [4649 kB]
Get:18 http://ports.ubuntu.com/ubuntu-ports noble-security/main arm64 Packages [2026 kB]
Fetched 39.1 MB in 7s (5829 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
4 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ca-certificates krb5-locales libbrotli1 libcurl4t64 libgssapi-krb5-2 libk5crypto3 libkeyutils1 libkrb5-3 libkrb5support0
  libldap-common libldap2 libnghttp2-14 libpsl5t64 librtmp1 libsasl2-2 libsasl2-modules libsasl2-modules-db libssh-4
  openssl publicsuffix
Suggested packages:
  krb5-doc krb5-user libsasl2-modules-gssapi-mit | libsasl2-modules-gssapi-heimdal libsasl2-modules-ldap
  libsasl2-modules-otp libsasl2-modules-sql
The following NEW packages will be installed:
  ca-certificates curl krb5-locales libbrotli1 libcurl4t64 libgssapi-krb5-2 libk5crypto3 libkeyutils1 libkrb5-3
  libkrb5support0 libldap-common libldap2 libnghttp2-14 libpsl5t64 librtmp1 libsasl2-2 libsasl2-modules
  libsasl2-modules-db libssh-4 openssl publicsuffix
0 upgraded, 21 newly installed, 0 to remove and 4 not upgraded.
Need to get 3560 kB of archives.
After this operation, 9899 kB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 openssl arm64 3.0.13-0ubuntu3.7 [985 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 ca-certificates all 20240203 [159 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 krb5-locales all 1.20.1-6ubuntu2.6 [14.8 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libkrb5support0 arm64 1.20.1-6ubuntu2.6 [34.8 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libk5crypto3 arm64 1.20.1-6ubuntu2.6 [85.8 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libkeyutils1 arm64 1.6.3-3build1 [9654 B]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libkrb5-3 arm64 1.20.1-6ubuntu2.6 [349 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libgssapi-krb5-2 arm64 1.20.1-6ubuntu2.6 [142 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libnghttp2-14 arm64 1.59.0-1ubuntu0.2 [74.5 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libpsl5t64 arm64 0.21.2-1.1build1 [57.4 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 publicsuffix all 20231001.0357-0.1 [129 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libbrotli1 arm64 1.1.0-2build2 [339 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libsasl2-modules-db arm64 2.1.28+dfsg1-5ubuntu3.1 [21.6 kB]
Get:14 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libsasl2-2 arm64 2.1.28+dfsg1-5ubuntu3.1 [54.7 kB]
```

```
Processing triggers for ca-certificates (20240203) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@b8dc6d541e31:/# curl --version
curl 8.5.0 (aarch64-unknown-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.
21.2 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.8
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs sm
tp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threa
dsafe TLS-SRP UnixSockets zstd
root@b8dc6d541e31:/# exit
exit
```

3. Запуск веб-сервера

```
exit
flocktory@MacBook-Air-Flocktory-69 ~ % docker run -d -p 8080:80 --name web-server nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
b15a4549c6e6: Pull complete
96522e0e3029: Pull complete
26a9266e72a1: Pull complete
1595112a18c2: Pull complete
c0700f204f65: Pull complete
21c8549f3077: Pull complete
d8ad8cd72600: Pull complete
0855a7fcacab: Pull complete
763e574ab084: Download complete
03a697649db4: Download complete
Digest: sha256:f46cb72c7df02710e693e863a983ac42f6a9579058a59a35f1ae36c9958e4ce0
Status: Downloaded newer image for nginx:alpine
9af46b05f5b58f0c4a14ce50100f32519b817ad50f106e9a05fa0e00b03593e5
```




```
9af46b05f5b58f0c4a14ce50100f32519b817ad50f106e9a05fa0e00b03593e5
flocktory@MacBook-Air-Flocktory-69 ~ % docker logs web-server
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/12 15:30:04 [notice] 1#1: using the "epoll" event method
2026/03/12 15:30:04 [notice] 1#1: nginx/1.29.6
2026/03/12 15:30:04 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/03/12 15:30:04 [notice] 1#1: OS: Linux 6.12.72-linuxkit
2026/03/12 15:30:04 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/12 15:30:04 [notice] 1#1: start worker processes
2026/03/12 15:30:04 [notice] 1#1: start worker process 30
2026/03/12 15:30:04 [notice] 1#1: start worker process 31
2026/03/12 15:30:04 [notice] 1#1: start worker process 32
2026/03/12 15:30:04 [notice] 1#1: start worker process 33
2026/03/12 15:30:04 [notice] 1#1: start worker process 34
2026/03/12 15:30:04 [notice] 1#1: start worker process 35
2026/03/12 15:30:04 [notice] 1#1: start worker process 36
```

```
192.168.65.1 - - [12/Mar/2026:15:31:53 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Mac
intosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-"
flocktory@MacBook-Air-Flocktory-69 ~ % docker exec -it web-server sh
/ # docker ps
```

4. Управление контейнерами

```
flocktory@MacBook-Air-Flocktory-69 ~ % docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
    NAMES
9af46b05f5b5   nginx:alpine   "/docker-entrypoint.…"   8 minutes ago   Up 8 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tc
p   web-server
flocktory@MacBook-Air-Flocktory-69 ~ % docker ps -a

CONTAINER ID   IMAGE          COMMAND                  CREATED             STATUS                          PORTS
                        NAMES
9af46b05f5b5   nginx:alpine   "/docker-entrypoint.…"   8 minutes ago       Up 8 minutes                    0.0.0.0:8080->80/t
cp, [::]:8080->80/tcp   web-server
b8dc6d541e31   ubuntu         "bash"                   About an hour ago   Exited (0) 9 minutes ago
                        exciting_lichterman
2c08209d64b3   hello-world    "/hello"                 About an hour ago   Exited (0) About an hour ago
                        eager_swirles
flocktory@MacBook-Air-Flocktory-69 ~ % docker stop web-server
web-server
flocktory@MacBook-Air-Flocktory-69 ~ % docker start web-server


web-server
flocktory@MacBook-Air-Flocktory-69 ~ % docker rm web-server
Error response from daemon: cannot remove container "web-server": container is running: stop the container before removing o
r force remove
flocktory@MacBook-Air-Flocktory-69 ~ % docker stop web-server
web-server
flocktory@MacBook-Air-Flocktory-69 ~ % docker rm web-server
web-server
flocktory@MacBook-Air-Flocktory-69 ~ % docker rmi nginx:alpine
Untagged: nginx:alpine
Deleted: sha256:f46cb72c7df02710e693e863a983ac42f6a9579058a59a35f1ae36c9958e4ce0
```

5. Работа с томами

```
Last login: Thu Mar 12 21:56:15 on ttys000
flocktory@MacBook-Air-Flocktory-69 ~ % docker volume create my-volume
my-volume
flocktory@MacBook-Air-Flocktory-69 ~ % docker run -it --name volume -test -d -v my-volume:/data ubuntu bash
43750eaa24ae24ecc99b18447cd85c6c8e01373b57cd04066af63db741ce1a27
flocktory@MacBook-Air-Flocktory-69 ~ % docker exec -it volume-test bash
root@6ea950e36241:/# echo "Hello from volume" > /data/test.txt
root@6ea950e36241:/# cat /data/test.txt
Hello from volume
root@6ea950e36241:/# exit
exit
```

```
2c08209d64b3   hello-world   "/hello"   26 hours ago     Exited (0) 26 hours ago              eager_swirles
flocktory@MacBook-Air-Flocktory-69 ~ % docker volume create my-volume
my-volume
flocktory@MacBook-Air-Flocktory-69 ~ % docker run -it --name volume-test4 -d -v my-volume:/data ubuntu bash
6b6e6f01d2fb3bae2a01518068e70b2b52f6ff88fbc02545ba22d4a45f332b22
flocktory@MacBook-Air-Flocktory-69 ~ % docker stop volume-test4
volume-test4
flocktory@MacBook-Air-Flocktory-69 ~ % docker rm volume-test4
volume-test4
flocktory@MacBook-Air-Flocktory-69 ~ % docker rm my-volume
Error response from daemon: No such container: my-volume
flocktory@MacBook-Air-Flocktory-69 ~ % docker volume ls
DRIVER    VOLUME NAME
local     my-volume
flocktory@MacBook-Air-Flocktory-69 ~ % docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS                     PORTS     NAMES
44b1f21312a4   ubuntu        "bash"     21 minutes ago   Up 21 minutes                        volume-test2
43750eaa24ae   ubuntu        "bash"     39 minutes ago   Up 39 minutes                        volume
b8dc6d541e31   ubuntu        "bash"     26 hours ago     Exited (0) 25 hours ago              exciting_lichterman
2c08209d64b3   hello-world   "/hello"   26 hours ago     Exited (0) 26 hours ago              eager_swirles
flocktory@MacBook-Air-Flocktory-69 ~ %
```